SECTION A – 2 MARKS

Q1. (a)

Statistics is the science of collecting, presenting, analysing and interpreting numerical data. In the words of Croxton and Cowden, statistics may be defined as "the collection, presentation, analysis and interpretation of numerical data".

Q1. (b)

1. **Primary data:** Data collected for the first time by the investigator himself, for a specific purpose. They are original in nature.
2. **Secondary data:** Data already collected by someone else and used by the investigator, such as published reports and records.

Q1. (c)

Tabulation is the systematic arrangement of classified data in rows and columns. It presents a large mass of data in a compact and orderly form, so that it is easy to understand and compare.

Q1. (d)

1. The sum of the deviations of the items from the arithmetic mean is always zero: $\Sigma(x - \bar{x}) = 0$.
2. The sum of the squares of the deviations from the arithmetic mean is the least: $\Sigma(x - \bar{x})^2$ is a minimum.

Q1. (e)

Largest value L = 100, smallest value S = 20.
Range = L − S = 100 − 20 = **80**.
Coefficient of range = (L − S) / (L + S) = 80 / 120 = **0.67**.

Q1. (f)

Standard deviation is the square root of the arithmetic mean of the squared deviations of the items from their arithmetic mean. It is denoted by σ: $\sigma = \sqrt{(\Sigma(x - \bar{x})^2 / N)}$. It is the most widely used measure of dispersion.

Q1. (g)

Correlation is the relationship between two or more variables, such that a change in one variable is accompanied by a change in the other. For example, price and demand, or income and expenditure, are correlated.

Q1. (h)

Regression is the statistical method of estimating the unknown value of one variable from the known value of another variable. It shows the average relationship between two variables in the form of an equation.

Q1. (i)

Classification is the process of arranging the collected data into groups or classes according to their common characteristics, such as area, time, attribute or size.

Q1. (j)

A **histogram** is a graph of a frequency distribution in which the class intervals are shown on the X-axis and the frequencies on the Y-axis. Each class is shown by a rectangle whose height is its frequency, and the rectangles are drawn next to one another.

SECTION B – 5 MARKS

Q2.

1. **Geographical classification:** Data are grouped by place or area, such as country, state, district or city, for example the population of each state.
2. **Chronological classification:** Data are grouped by time, such as years, months or weeks, for example the sales of a company year by year.
3. **Qualitative classification:** Data are grouped by an attribute that cannot be measured, such as sex, religion, literacy or occupation.
4. **Simple and manifold classification:** In simple classification data are divided by one attribute into two classes (for example literate and illiterate); in manifold classification they are divided by two or more attributes at a time.
5. **Quantitative classification:** Data are grouped by a characteristic that can be measured, such as height, weight, income or marks, using class intervals. Such a grouping is a frequency distribution.

Q3.

1. **Direct personal investigation:** The investigator collects the information personally from the persons concerned. It suits a small, local enquiry and gives accurate data.
2. **Indirect oral investigation:** Information is collected from third persons (witnesses) who know the facts, as in enquiries by committees and commissions.
3. **Information through local correspondents:** Local agents or correspondents are appointed in different areas, and they send the information regularly, as newspapers do.
4. **Mailed questionnaire:** A list of questions is sent by post to the informants, who fill it in and return it. It covers a wide area at low cost.
5. **Schedules sent through enumerators:** Trained enumerators take the schedule to the informants, ask the questions and fill in the answers themselves, as in the census.

Q4.

x	f	cf
20	25	25
60	48	73
80	60	133
100	57	190
120	29	219
140	10	229
160	8	237
200	2	239
	Σf = N = 239	

Q1 = size of [(N + 1) / 4]th item = (239 + 1) / 4 = 240 / 4 = 60th item. The 60th item falls in cf 73, so **Q1 = 60**.
Q3 = size of 3[(N + 1) / 4]th item = 3 (240 / 4) = 3 (60) = 180th item. The 180th item falls in cf 190, so **Q3 = 100**.
Quartile deviation = (Q3 − Q1) / 2 = (100 − 60) / 2 = 40 / 2 = **20**
Coefficient of QD = (Q3 − Q1) / (Q3 + Q1) = (100 − 60) / (100 + 60) = 40 / 160 = **0.25**

Q5.

x	log x
20	1.3010
350	2.5441
450	2.6532
600	2.7782
861	2.9350
958	2.9814
n = 6	**Σ log x = 15.1929**

GM = Antilog (Σ log x / n) = Antilog (15.1929 / 6) = Antilog 2.5322
GM = 10^2.5322 = **340.52**

Q6.

Table showing the distribution of employees according to class, sex and salary grade

Salary (Rs.)	Sup. M	Sup. F	Asst. M	Asst. F	Clerk M	Clerk F	Total
Below 3,000							
3,000 – 5,000							
5,000 and above							
Total							

Sup. = Supervisor, Asst. = Assistant, M = Male, F = Female. Each cell holds the number of employees.
Source: ..... **Note:** .....

Q7.

N = 20, $\bar{x}$ = 4, $\bar{y}$ = 2, Σdx² = 1,680, Σdy² = 320, Σdxdy = 480
bxy = Σdxdy / Σdy² = 480 / 320 = **1.5**
byx = Σdxdy / Σdx² = 480 / 1,680 = 0.2857 ≈ **0.29**

x on y

$x - \bar{x} = bxy\ (y - \bar{y})$
x − 4 = 1.5 (y − 2)
x − 4 = 1.5y − 3
x = 1.5y + 1

y on x

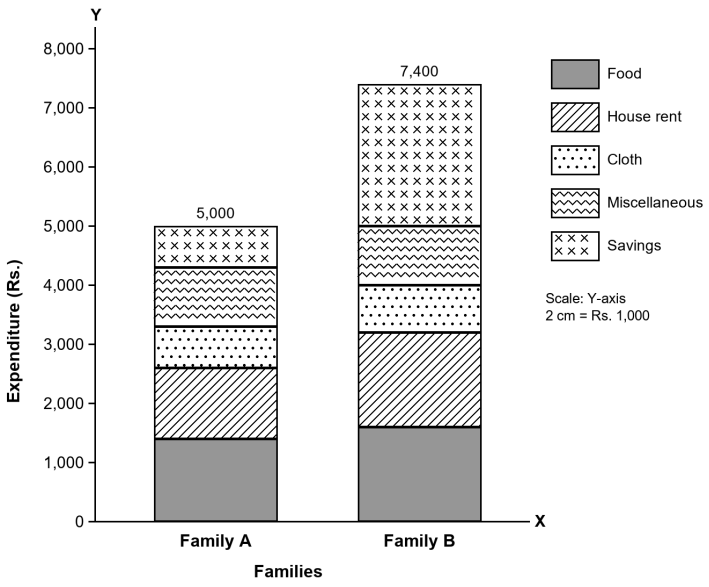
$y - \bar{y} = byx\ (x - \bar{x})$
y − 2 = 0.29 (x − 4)
y − 2 = 0.29x − 1.16
y = 0.29x + 0.84

When y = 7

x = 1.5 (7) + 1 = 10.5 + 1 = **11.5**

Q8.

Item	Family A	Family B
Food	1,400	1,600
House rent	1,200	1,600
Cloth	700	800
Miscellaneous	1,000	1,000
Savings	700	2,400
Total	**5,000**	**7,400**



Q9.

Frequency distribution of the given data

Class	Tally bars	Frequency
0 – 10	IIII	4
10 – 20	𝍸 III	8
20 – 30	𝍸 𝍸	10
30 – 40	IIII	4
40 – 50	IIII	4
	Total	**30**

SECTION C – 10 MARKS

Q10.

A = 14.5, c = 10, d' = (x − A) / c; exclusive classes (9.5 – 19.5 …) for the median.

CI	f	x	d'	fd'	cf
10 – 19	10	14.5	0	0	10
20 – 29	18	24.5	1	18	28
30 – 39	8	34.5	2	16	36
40 – 49	6	44.5	3	18	42
50 – 59	4	54.5	4	16	46
60 – 69	2	64.5	5	10	48
70 – 79	5	74.5	6	30	53
80 – 89	7	84.5	7	49	60
	N = 60			**Σfd' = 157**	

Mean

$\bar{x} = A + (\Sigma fd' / N) \times c = 14.5 + (157 / 60) \times 10 = 14.5 + 2.6167 \times 10 = 14.5 + 26.17 =$ **40.67**

Median

m = N / 2 = 60 / 2 = 30th item → class 29.5 – 39.5 (cf 36).
L1 = 29.5, L2 = 39.5, f = 8, c (cf of the class before) = 28
Median = L1 + ((L2 − L1) / f) (m − c) = 29.5 + ((39.5 − 29.5) / 8) (30 − 28) = 29.5 + 1.25 (2) = 29.5 + 2.5 = **32**

Mode

$Z = 3M - 2\bar{x} = 3\ (32) - 2\ (40.67) = 96 - 81.33 =$ **14.67**
Mean = 40.67, Median = 32, Mode = 14.67

Q11.

A = 15, c = 10, d' = (x − A) / c

CI	f	x	d'	fd'	d'²	fd'²	fx
10 – 20	10	15	0	0	0	0	150
20 – 30	18	25	1	18	1	18	450
30 – 40	20	35	2	40	4	80	700
40 – 50	6	45	3	18	9	54	270
50 – 60	4	55	4	16	16	64	220
60 – 70	2	65	5	10	25	50	130
	N = 60			**Σfd' = 102**		**Σfd'² = 266**	**Σfx = 1,920**

Standard deviation

$\sigma = \sqrt{(\Sigma fd'^2 / N - (\Sigma fd' / N)^2)} \times c = \sqrt{(266 / 60 - (102 / 60)^2)} \times 10$
$\sigma = \sqrt{(4.4333 - (1.7)^2)} \times 10 = \sqrt{(4.4333 - 2.89)} \times 10 = \sqrt{1.5433} \times 10 = 1.2423 \times 10 =$ **12.42**

Mean

$\bar{x} = \Sigma fx / N = 1,920 / 60 =$ **32**

Coefficient of variation

$CV = (\sigma / \bar{x}) \times 100 = (12.42 / 32) \times 100 = 0.3882 \times 100 =$ **38.82%**

Q12.

$\bar{x} = 141 / 8 = 17.6$, $\bar{y} = 286 / 8 = 35.7$, $x = X - \bar{x}$, $y = Y - \bar{y}$

X	Y	x	y	xy	x²	y²
8	25	−9.6	−10.7	102.72	92.16	114.49
10	30	−7.6	−5.7	43.32	57.76	32.49
15	32	−2.6	−3.7	9.62	6.76	13.69
17	35	−0.6	−0.7	0.42	0.36	0.49
20	37	2.4	1.3	3.12	5.76	1.69
22	40	4.4	4.3	18.92	19.36	18.49
24	42	6.4	6.3	40.32	40.96	39.69
25	45	7.4	9.3	68.82	54.76	86.49
141	**286**			**Σxy = 287.26**	**Σx² = 277.88**	**Σy² = 307.52**

Coefficient of correlation

$r = \Sigma xy / \sqrt{(\Sigma x^2 \times \Sigma y^2)} = 287.26 / \sqrt{(277.88 \times 307.52)} = 287.26 / \sqrt{85,453.66} = 287.26 / 292.32 =$ **0.983**

Probable error

$PE = 0.6745 \times (1 - r^2) / \sqrt{n} = 0.6745 \times (1 - 0.9657) / \sqrt{8} = 0.6745 \times 0.0343 / 2.828 = 0.6745 \times 0.0121 =$ **0.0082**

r = 0.983, PE = 0.0082. Very high positive correlation; r > 6 PE, so it is significant.